Day 2 — Java Programming Fundamentals

Answers and Solutions

1. Introduction to Java
Java is a high-level, object-oriented programming language. Java programs are compiled into bytecode, which can run on a Java Virtual Machine (JVM).

2. JDK, JRE, and JVM
JDK: Java Development Kit; tools used to develop and compile Java programs.
JRE: Java Runtime Environment; components needed to run Java applications.
JVM: Java Virtual Machine; executes Java bytecode.

3. Variables and data types
A variable stores a value. Common primitive types include `byte`, `short`, `int`, `long`, `float`, `double`, `char`, `boolean`. Reference types include classes, arrays, and strings.

4. Operators
Arithmetic: `+ - * / %`; comparison: `== != > < >= <=`; logical: `&& || !`; assignment: `= += -= *=`; increment/decrement: `++ --`.

5. Conditional statements
Use `if`, `else if`, and `else` to make decisions. Use `switch` when selecting among cases.

6. Loops
`for` is useful when the iteration count is known; `while` checks the condition before each iteration; `do-while` runs its body at least once.

7. Input and output
`System.out.println()` prints output. `Scanner` can read input from the keyboard.

8. Basic problem-solving programs

The following examples can each be saved in a file named `Main.java` and run separately. Each program uses the class name `Main`.

A. Check whether a number is even or odd

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        if (n % 2 == 0) {
            System.out.println("Even");
        } else {
            System.out.println("Odd");
        }
        sc.close();
    }
}
```

B. Find the largest of three numbers

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter three numbers: ");
        int a = sc.nextInt();
        int b = sc.nextInt();
        int c = sc.nextInt();

        if (a >= b && a >= c) {
            System.out.println("Largest: " + a);
        } else if (b >= a && b >= c) {
```

```
            System.out.println("Largest: " + b);
        } else {
            System.out.println("Largest: " + c);
        }
        sc.close();
    }
}
```

C. Calculate factorial

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a non-negative integer: ");
        int n = sc.nextInt();

        if (n < 0) {
            System.out.println("Factorial is not defined for negative integers.");
        } else {
            long fact = 1;
            for (int i = 1; i <= n; i++) {
                fact *= i;
            }
            System.out.println("Factorial: " + fact);
        }
        sc.close();
    }
}
```

D. Check whether a number is prime

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter an integer: ");
        int n = sc.nextInt();

        boolean prime = n >= 2;
        for (int i = 2; i <= n / i && prime; i++) {
            if (n % i == 0) {
                prime = false;
            }
        }

        System.out.println(prime ? "Prime" : "Not prime");
        sc.close();
    }
}
```

E. Reverse a number

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter an integer: ");
        int n = sc.nextInt();

        int temp = Math.abs(n);
        long reverse = 0;
        while (temp > 0) {
            int digit = temp % 10;
            reverse = reverse * 10 + digit;
            temp /= 10;
        }
        if (n < 0) reverse = -reverse;

        System.out.println("Reversed: " + reverse);
        sc.close();
    }
}
```

F. Print a multiplication table

```
import java.util.Scanner;
```

```
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        for (int i = 1; i <= 10; i++) {
            System.out.println(n + " x " + i + " = " + (n * i));
        }
        sc.close();
    }
}
```

9. Configure the Java environment

1. Install a JDK from a trusted Java distribution. 2. Open a terminal/Command Prompt and run `java -version` and `javac -version`. 3. If both commands show versions, Java and the compiler are available on PATH. 4. Create `Main.java`, compile it using `javac Main.java`, then run it using `java Main`. In an IDE such as IntelliJ IDEA, select an installed JDK in Project Structure / SDK settings.

10. Expected output checklist

✓ Java JDK installed and configured.
✓ Programs written and run successfully.
✓ Tested even/odd, largest of three, factorial, prime, reverse number, and multiplication table.

Note: Factorial values above 20 overflow a Java long in this simple program. The reverse-number example uses int input, so the absolute value of Integer.MIN_VALUE is a special edge case.